

1 Characterization of the amylase enzyme

EXPERIMENT 1

Kinetic characterization of an enzyme

Goal

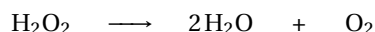
The goal of this experiment is to use the initial rate method to obtain the Michaelis-Menten constants for the starch hydrolysis catalyzed by amylase.

Background

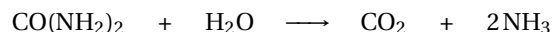
Enzymes are biological catalysts responsible for the speeding up of biochemical reactions. *Catalase* or *amylase* are two enzymes examples; the former is responsible for breaking down hydrogen peroxide (H_2O_2) whereas the latter catalyzes the hydrolysis of starch. Enzymes are particularly relevant in biochemistry accelerating chemical reactions at rates necessary for life. Most biochemical reactions require an enzyme. This section covers the chemical kinetic principles behind biological catalysis and the factors that impact enzyme activity.

What are enzymes

Enzymes are large biological molecules, in general proteins, able to speed up chemical reactions. Enzymes are more efficient than regular catalysts. The activation energy of enzyme-catalyzed chemical reactions tends to be smaller than that of regularly-catalyzed reactions. For example, the activation energy of the Fe^{2+} -catalyzed decomposition of hydrogen peroxide

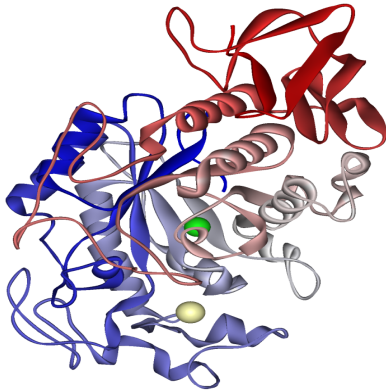


is 42 KJ/mol, whereas that for the catalase-catalyzed decomposition is only 7 KJ/mol. As enzymes are large molecules, only a small part of the molecule called the active site is actively involved in the catalytic process. Enzymes are associated with coenzymes which are inorganic compounds essential for catalyst action. On one hand, we can find very specific catalysts. An example would be urease which only catalyzes the hydrolysis of urea ($\text{CO}(\text{NH}_2)_2$), the reaction between urea and water:



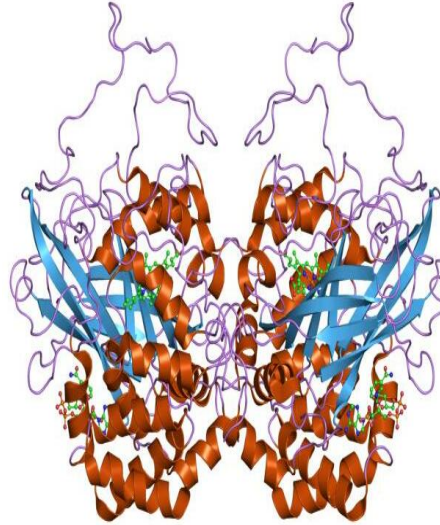
On the other hand, we can also find less specific enzymes such as proteolytic enzymes which catalyze the hydrolysis of the peptide linkage that connects amino acids in proteins. In fact, these types of enzymes are group-specific and will catalyze the hydrolysis of the peptide linkage in any molecule with such a chemical bond.

▼Amylase



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▼Catalase



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Amylase

Amylase is a digestive enzyme involved in the hydrolysis of complex carbohydrates (like starch) into simpler sugars. It is primarily produced by the pancreas and salivary glands aiding digestion. It is present in human saliva, and it was named after the Ancient Greek name for saliva. It exists in three different forms: α -amylase, β -amylase, and γ -amylase. α -amylase is the major form of amylase found in humans and other mammals, also being the fastest. β -amylase is present in plants and microbes, being responsible for the ripening of fruit as it breaks starch into maltose, resulting in the sweet flavor of ripe fruit. γ -amylase, present in animals and microbes, is most active around pH 3, being the most acidic optimum pH of all amylases.

Michaelis-Menten equation

The Michaelis-Menten equation gives the rate of an enzyme-catalyzed reaction in terms of the concentration of the enzyme.

$$v = \frac{V_m[S]}{K_m + [S]} \quad (1)$$

where:

v is the enzyme-catalyzed reaction rate

$[S]$ is the enzyme concentration

V_m is the maximal velocity

K_m Michaelis constant

It involves two different constants V_m which is the maximal rate and K_m , the Michaelis constant. Enzymes speed up chemical rate. However, when the enzyme concentration is too large, the enzyme catalyzed rate reaches a plateau with a maximal rate. At this point increasing the enzyme concentration does not increase the reaction rate. K_m is a constant that involves the reaction rate constant of the enzyme-catalyzed reaction mechanism.

Lineweaver-Burk linear equation

The Michaelis-Menten equation is not a linear relationship and hence it is not suitable to fit experimental data. Linear relations are suitable for obtaining kinetic data because they are simple to analyze when dealing with large datasets. Perhaps more importantly, errors are well-understood in linear expressions. The quality of the linear fit simply relates to the correlation coefficient.

Hence, the Michaelis-Menten equation can be recast into a first linear expression by simply using reciprocals. The resulting equation in which the inverse of the rate $1/v$ (vertical axis) is plotted against the inverse of $1/[S]$ (horizontal axis) is called the Lineweaver-Burk equation

$$\frac{1}{v} = \frac{K_m}{V} \cdot \frac{1}{[S]} + \frac{1}{V} \quad (2)$$

where:

$1/v$ is the inverse of the enzyme-catalyzed reaction rate

$1/[S]$ is the inverse of the enzyme concentration

V_m is the maximal velocity

K_m Michaelis constant

The slope of this plot is negative. From the intercept we can obtain V_m and from the slope we can calculate K_m .

Eadie linear equation

The Michaelis-Menten equation can be recast into a second linear expression as shown below:

$$v = V_m - K_m \cdot \frac{v}{[S]} \quad (3)$$

is called the Eadie equation. In the Eadie equation, the rate $1/v$ (vertical axis) is plotted against $v/[S]$ (horizontal axis). The slope of this plot is negative. The advantage of this equation is that it spreads more data than the Lineweaver-Burk equation.

PH and temperature effects

Enzyme activity depends on pH and temperature. pH has a strong impact on enzyme activity passing through a maximum as pH increases. Similarly, enzyme activity also passes through a maximum as temperature increases. Enzymes show maximum activity at an *optimum pH* and *optimum temperature*. This maximum depends on the nature of the enzyme and the catalyzed reaction. They reversibly lose activity at pH and temperature slightly lower or higher than their optimum. Enzyme deactivation is irreversible if the acidity or basicity is too high or if temperature is too high or low. One explanation for the pH effects is that ionization of protein carboxylic and amino groups occurs when pH is not taken too far within the optimum pH. Differently, enzymes lose their tertiary structure hence losing their activity at very low or high pH values or temperature higher than 30°C.

Because of their shape, graphs displaying catalyst activity vs pH or temperature are often referred to as volcano plots. Figure ?? displays pH and temperature volcano plots for the activity of amylase.

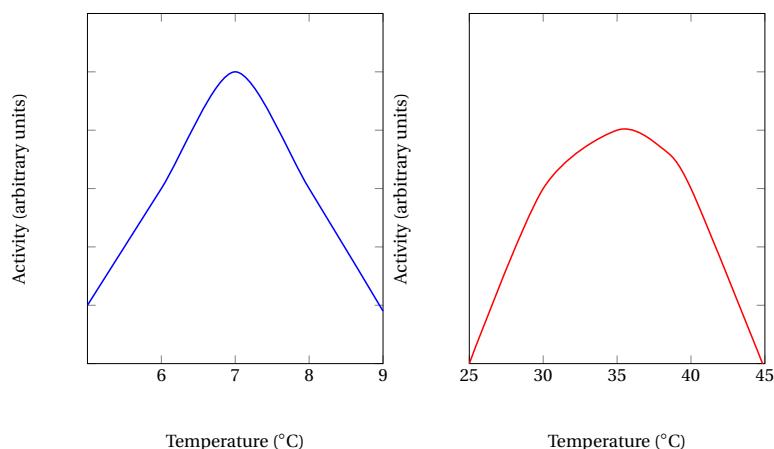


Figure ?? Schematic representation of pH (left) and temperature (right) volcano plots for the activity of amylase. Amylase is the digestive enzyme that breaks down complex carbohydrates (like starch) into simpler sugars, aiding digestion. The optimum temperature is 37°C, and the optimum pH range is 6.7-7.0.

The experiment

This experiment is divided into a series of parts. First, you will calibrate the spectrophotometer to transform absorbances into concentrations; you will do this by measuring absorbances at decreasing starch concentrations. Then you will measure the initial velocity v_0 for the amylase-catalyzed starch hydrolysis for high starch concentrations, making sure these measurements are accurate and reproducible. Afterwards, you will measure a set of v_0 for decreasing starch concentrations, to calculate the Michaelis-Menten constants K_m and V_m . To obtain these constants you will use two different linearizations of the Michaelis-Menten equation, building the Lineweaver-Burk and the Eadie-Hofstee plot. Finally, you will study the impact of PH on enzyme activity by measuring the initial rate for the reaction at different PH values.

Part A. Determining Lambert-Beer's constant $\epsilon \cdot l$.

Starch is the substrate of the enzyme-catalyzed reaction studies in this experiment. To transform Absorbances into molarity you will need the Lambert-Beer's constant. You will measure this constant by measuring Absorbances for a set of starch dilutions. You will take advantage of the fact that starch reacts with Iodine (I_2) in the presence of Iodide ion (I^-) to form an intensely colored blue complex, which is visible at very low concentrations of Iodine, making it an excellent indicator. The intensity increases linearly with the starch concentration.

Parts B. Determination and validation of initial velocities

You will measure the initial velocities of the reaction by starting the enzyme-catalyzed reaction and measure how the mixture's absorbance changes for a short time, just a minute; measuring absorbance every 15 seconds with a stopwatch. Using a straight part of the A vs. t graph and a linear regression, using Lambert-Beer's constant to transform absorbances into concentration, the slope of the plot corresponds to the reaction's initial velocity. You will repeat your measurement a total of three times to assess the accuracy and reproducibility of your rate measurements.

Part C. Determination of the Michaelis-Menten constants

To calculate the Michaelis-Menten constants you will measure initial reaction rates (v_0) for decreasing starch concentrations. As the rate of an enzyme-catalyzed reaction depends on the substrate concentration ($[S]$) using the Michaelis-Menten equation, graphing v_0 vs. $[S]$ (this is the first kinetic plot) will give a hyperbolic plot, not suitable to obtain the Michaelis-Menten constants. You will construct two more kinetic plots: the double-reciprocal Lineweaver-Burk graph by plotting $1/v_0$ vs. $1/[S]$ (this is the second kinetic plot) and the improved Eadie-Hofstee graph by plotting v_0 vs. $v_0/[S]$ (this is the third kinetic plot). The slope and intercept of both plots will be used to obtain K_m and V_m , the two Michaelis-Menten constants.

Part D. PH effect on enzyme activity

To estimate the PH effect on enzyme activity you will measure initial velocities after adding increasing quantities of an acid and a base. You will use PH paper to measure the PH of the starch and enzyme. Then you will add some drops of an acid, measure the initial velocity, and then measure its PH. Finally, you will add some drops of a base, measure the initial velocity, and then measure its PH. The initial velocity relates to enzyme activity. By measuring how this property changes with PH you will be able to estimate the optimum PH of the enzyme.

Procedure

Part A-1. Starch serial dilution

- ☐ Step 1: – In the following you will prepare a series of starch dilutions with concentrations: 2%, 1%, 0.5%, 0.25%, and 0.125%.
- ☐ Step 2: – Find five jumbo test tubes with stoppers and label them as A, B, C, D, and E.
- ☐ Step 3: – Pipet 5ml of distilled water into test tubes B, C, D, and E.

- ☐ *Step 4:* – Obtain about 25 mL of 2% starch in a small beaker. Make sure you do not contaminate the bottle of reagent by always retrieving out of the bottle into a beaker and never directly pipetting from the bottle. Clean and rinse a 10mL Mohr's pipette with the enzyme solution.
- ☐ *Step 5:* – Using the Mohr's pipet, pipet 5ml of starch into test tube A (test tube A is the starch 2% solution) and 5ml into test tube B.
- ☐ *Step 6:* – Pipet 5ml of starch into test tube A and 1ml into test tube B. Close test tube B with a stopper, and mix well the content of test tube B by inverting the test tube 3 times at constant speed. Test tube B is the starch 1% solution.
- ☐ *Step 7:* – Pipet 5ml of test tube B into test tube C. Close test tube C with a stopper, and mix well its content by inverting the tube 3 times at constant speed. Test tube C is the starch 0.5% solution.
- ☐ *Step 8:* – Pipet 5ml of test tube C into test tube D. Close test tube D with a stopper, and mix well its content by inverting the tube 3 times at constant speed. Test tube D is the starch 0.25% solution.
- ☐ *Step 9:* – Pipet 5ml of test tube D into test tube E. Close test tube E with a stopper, and mix well its content by inverting the tube 3 times at constant speed. Test tube E is the starch 0.125% solution.
- ☐ *Step 10:* – Add 1 drop of the I_2-I^- solution into each test and mix all test tubes well until the color is homogeneous. With your serial dilution done properly, you should see test tube A darker in comparison to the other test tubes with intensity decreasing from test tube A to E.

Part A-2. Starch standard calibration

- ☐ *Step 1:* – Start the spectrophotometer. The bulb needs time to heat up to a stable temperature.
- ☐ *Step 2:* – Get a 250 mL beaker for waste and fill a 100 mL beaker with distilled water.
- ☐ *Step 3:* – Using the plastic dropper, fill the cuvette 3/4 full with distilled water.
- ☐ *Step 4:* – Insert the cuvette into the spectrophotometer, set the wavelength to 540 nm , and press "blank" or "zero".
- ☐ *Step 5:* – Use a new cuvette. Fill the cuvette 3/4 full with the solution from test tube E.
- ☐ *Step 6:* – Insert the cuvette into the spectrophotometer to measure the absorbance. Record the result.
- ☐ *Step 7:* – Repeat for test tubes D, C, B, and A, always using new cuvettes for each measurement. Write down the results in the results section.
- ☐ *Step 8:* – Plot absorbance vs. concentration to obtain the Lambert-Beer's constant k .
- ☐ *Step 9:* – Do not proceed unless you show the instructor your graph.

Parts B. Determination and validation of initial velocities

- ☐ *Step 1:* – In the following you will start the enzyme-catalyzed reaction to measure initial velocities.
- ☐ *Step 2:* – Gather test tubes A-E with their stoppers while you get ready to start the reaction.
- ☐ *Step 3:* – Find three new jumbo test tubes with stoppers and label them as A1, A2, and A3. With a clean Mohr's pipette, pipette 2mL of stock 2% starch solution into test tubes A1, A2, and A3. Make sure your Mohr's pipette is clean.
- ☐ *Step 4:* – Obtain about 20 mL of the enzyme solution (this is the enzyme solution or the artificial saliva solution). Make sure you do not contaminate the bottle of reagent by always retrieving liquid out of the bottle into a beaker and never directly pipetting from the bottle. Clean and rinse a Mohr's pipette with the enzyme solution.

- ☐ *Step 5:* – Pipette 2mL of the enzyme solution into test tube A1 and start counting time. Mix well the content by inverting the tube 3 times at constant speed. Very quickly, using the plastic dropper, fill a cuvette 3/4 full with the mixture and start measuring the absorbance.
- ☐ *Step 6:* – Measure absorbance every 15 seconds for two minutes. Write down the results in the results section.
- ☐ *Step 7:* – Repeat the previous two steps for test tubes A2 and A3. Once done, clean the test tubes A2 and A3 and have them ready for reuse.
- ☐ *Step 8:* – Process the data in the results section and obtain initial velocities for your three measurements. Make sure your measured initial velocities are reproducible.

Part C. Determination of the Michaelis-Menten constants

- ☐ *Step 1:* – In the following you are going to measure enzyme activity at different starch concentrations [S].
- ☐ *Step 2:* – Again, gather test tubes A-E with their stoppers while you get ready to start the reaction.
- ☐ *Step 3:* – Find three new jumbo test tubes with stoppers and label them as B1, C1, and D1. With a clean Morh's pipette, pipette 2mL from test tube B into B1. Then pipette 2mL from test tube C into C1, and 2mL from test tube D into D1.
- ☐ *Step 4:* – Pipette 2mL of the enzyme solution into test tube B1 and start counting time. Mix well the content by inverting the tube 3 times at constant speed. Very quickly, using the plastic dropper, fill a cuvette 3/4 full with the mixture and start measuring the absorbance.
- ☐ *Step 5:* – Measure absorbance every 15 seconds for two minutes. Write down the results in the results section.
- ☐ *Step 6:* – Repeat the previous two steps for test tubes C1 and D1. Once done, clean the test tubes B1, C2, and D1 and have them ready for reuse.
- ☐ *Step 7:* – Process the data in the results section and obtain initial velocities for your three measurements.
- ☐ *Step 8:* – Graph a Michaelis-Menten plot plotting initial velocity v_0 (vertical axis) vs. starch concentration [S] (horizontal axis).
- ☐ *Step 9:* – Graph a Lineweaver-Burk plot plotting the inverse of initial velocity $1/v_0$ (vertical axis) vs. the inverse of starch concentration $1/[S]$ (horizontal axis).
- ☐ *Step 10:* – Graph a Eadie-Hofstee plot plotting the initial velocity v_0 (vertical axis) vs. initial velocity and starch concentration ratio $v_0/[S]$ (horizontal axis).
plots

Part D. PH effect on enzyme activity

- ☐ *Step 1:* – In the following you will measure enzyme activity at different PH.
- ☐ *Step 2:* – Again, gather test tubes A-E with their stoppers while you get ready to start the reaction.
- ☐ *Step 3:* – Find two new jumbo test tubes with stoppers and label them as A3, and A4. With a clean Morh's pipette, pipette 2mL of stock 2% starch solution into test tubes A3, and A4. Make sure your Morh's pipette is clean.
- ☐ *Step 4:* – Add 5 drops of 2M HCl to test tube A3. Close the test tube with a stopper, and mix well its content by inverting the test tube 3 times at constant speed. Measure its PH using PH paper and write down the value in the results section.

- ☐ *Step 5:* – Add 5 drops of 2M NaOH to test tube A4. Close the test tube with a stopper, and mix well its content by inverting the test tube 3 times at constant speed. Measure its PH using PH paper and write down the value in the results section.
- ☐ *Step 6:* – Pipette 2mL of the enzyme solution into test tube A3 and start counting time. Mix well the content by inverting the tube 3 times at constant speed. Very quickly, using the plastic dropper, fill a cuvette 3/4 full with the mixture and start measuring the absorbance.
- ☐ *Step 7:* – Measure absorbance every 15 seconds for two minutes. Write down the results in the results section.
- ☐ *Step 8:* – Repeat the previous two steps for test tube A4.
- ☐ *Step 9:* – Graph a volcano plot plotting initial velocity v_0 (vertical axis) vs. PH (horizontal axis).

Calculations

- ① This is lab temperature. You will use this value for Part B as well.
- ① This is the volume of iodine added in each experiment.
- ② This is the volume of thiosulfate added in each experiment.
- ③ This is the volume of peroxydisulfate added in each experiment.
- ④ This is the total volume v_{total} of the reacting mixture.
- ⑤ This is the peroxydisulfate concentration in the mixture:

$$[S_2O_8^{-2}] = \frac{v_{S_2O_8^{-2}} \cdot c_{S_2O_8^{-2}}}{v_{total}} = \frac{③ \cdot 0.2}{④}$$

- ⑥ This is the iodine concentration in the mixture:

$$[I^-] = \frac{v_{I^-} \cdot c_{I^-}}{v_{total}} = \frac{① \cdot 0.2}{④}$$

- ⑦ This is the change in thiosulfate concentration during the reaction:

$$\Delta[S_2O_3^{-2}] = \frac{v_{S_2O_3^{-2}} \cdot c_{S_2O_3^{-2}}}{v_{total}} = \frac{② \cdot 0.01}{④}$$

- ⑧ This is the average time Δt for each experiment.

- ⑨ This is the reaction rate:

$$r = \frac{\Delta[S_2O_3^{-2}]}{2 \cdot \Delta t} = \frac{⑦}{2 \cdot ⑧}$$

- ⑩ This is the peroxydisulfate order in the rate law:

$$a = \frac{\ln\left(\frac{r_{Exp1}}{r_{Exp2}}\right)}{\ln\left(\frac{[S_2O_8^{-2}]_{Exp1}}{[S_2O_8^{-2}]_{Exp2}}\right)} = \frac{\ln\left(\frac{(9)_{Exp1}}{(9)_{Exp2}}\right)}{\ln\left(\frac{(5)_{Exp1}}{(5)_{Exp2}}\right)}$$

- (11) This is the iodide order in the rate law:

$$b = \frac{\ln\left(\frac{r_{Exp1}}{r_{Exp3}}\right)}{\ln\left(\frac{[I^-]_{Exp1}}{[I^-]_{Exp3}}\right)} = \frac{\ln\left(\frac{(9)_{Exp1}}{(9)_{Exp3}}\right)}{\ln\left(\frac{(6)_{Exp1}}{(6)_{Exp3}}\right)}$$

- (12) This is the rate constant calculated for each experiment:

$$k = \frac{r}{[S_2O_8^{-2}] \cdot [I^-]} = \frac{(9)}{(5) \cdot (6)}$$

- (13) This is the average rate constant.

- (14) The data on this row should be the same as in the first row in the Results table of Part B.

- (15) This is the activation energy. You need to use the two different reaction constants at different temperatures. R is 3.184 J/K·mol:

$$E_a = \frac{T_1 \cdot T_2}{T_2 - T_1} \cdot R \cdot \ln\left(\frac{k_2}{k_1}\right)$$

STUDENT INFO

Name: _____ Date: _____

Pre-lab Questions

Kinetic characterization of an enzyme

1. What is the difference between a test tube and a jumbo test tube?.
2. For the reaction $2A + 3B \rightarrow C$ what is the mathematical expression that relates the rate of A (r_a) and the rate of B (r_b)?
3. Explain the Iodine clock reaction used in this experiment.
4. What is starch and where can you find it in nature?
5. Low-quality boiled ham often contains potatoes to make it cheaper. How could you test if one piece of ham is pure meat or it contains potatoes?
6. Find out the difference between a **Mohr pipette** and a **transfer pipette**.

STUDENT INFO

Name: _____ Date: _____

**Results
EXPERIMENT****Kinetic characterization of an enzyme**Part A. Starch calibration

Test tube #	[S] (% (w/v))	A
	①	②
A	2.000	
B	1.000	
C	0.500	
D	0.250	
E	0.125	
③	$k_{\lambda} =$	_____
④	$r^2 =$	_____

Parts B. Initial velocities

Test tube A		Test tube A#1		Test tube A#2	
A	t (s)	A	t (s)	A	t (s)
5	6	5	6	5	6
7 $v_0=$	_____	_____	_____	_____	_____
8 $r^2=$	_____	_____	_____	_____	_____

[illegible]

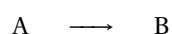
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Post-lab Questions

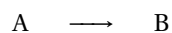
Kinetic characterization of an enzyme

1. Use the data below to calculate the rate law of the following reaction:



Experiment	r (M/s)	$[A]$, (M)
1	5×10^{-3}	0.5
2	5×10^{-3}	0.6

2. Use the data below to calculate the rate law of the following reaction:



Experiment	r (M/s)	$[A]$, (M)
1	0.5	0.253
2	1	0.320

3. Use the data below to calculate the rate law:

Experiment	$[A]$ (M)	$[B]$ (M)	r (M/s)
1	0.1	0.02	1.03×10^{-3}
2	0.1	0.04	2.08×10^{-3}
3	0.2	0.01	1.04×10^{-3}
4	0.4	0.01	2.08×10^{-3}

